FCC-ue and Muon Collider: $HH \to 4b$ and $\nu\bar{\nu}4b$ Rates

Preliminary LO parton-level cross sections with $P_{\mu^+} = +80\%$

Updated collider configurations. This comparison uses $\mu^+(5\,\text{TeV}) + e^-(183\,\text{GeV})$ at FCC-ue ($\sqrt{s} = 1.913\,\text{TeV}$) and $\mu^+(1\,\text{TeV}) + \mu^-(1\,\text{TeV})$ at the muon collider ($\sqrt{s} = 2.000\,\text{TeV}$).

Beam-energy configurations

Collider	Incoming beams	$\sqrt{s}$
FCC-ue (μ^+e^-)	$E_{\mu^+} = 5.0\,\text{TeV}$, $E_{e^-} = 183\,\text{GeV}$	1.913 TeV
Muon collider ($\mu^+\mu^-$)	$E_{\mu^+} = E_{\mu^-} = 1.0\,\text{TeV}$	2.000 TeV

Fiducial selection

- All four final-state $b/\bar{b}$ partons satisfy $|\eta_b^{\text{lab}}| < 5$.
- Form two *disjoint*, opposite-charge $b\bar{b}$ pairs. Accept the event if either distinct pairing has *both* masses in $110 \le m_{b\bar{b}}^{(1,2)} \le 130\,\text{GeV}$; count each event once.
- **No minimum p_T cut** is applied (in particular, the optional 5 GeV threshold is *not* used).
- These are parton-level estimates (no showering, jet clustering, tagging or detector simulation).

Cross sections before and after cuts

All entries are in fb. After-cuts errors are approximate Monte Carlo statistical uncertainties only.

Collider / beams	Final state	Before cuts	After cuts $\pm\delta_{\text{MC}}$
FCC-ue μ^+e^-	$\nu_e\bar{\nu}_\mu HH \to \nu_e\bar{\nu}_\mu\, 4b$	0.493162	**0.476106 ± 0.000788**
	$\nu_e\bar{\nu}_\mu\, 4b$	4.562030	**0.507450 ± 0.008324**
Muon collider $\mu^+\mu^-$	$\sum_{\ell=e,\mu,\tau} \nu_\ell\bar{\nu}_\ell\, HH(\to 4b)$	0.547571	**0.546063 ± 0.000850**
	$\sum_{\ell=e,\mu,\tau} \nu_\ell\bar{\nu}_\ell\, 4b$	4.929144	**0.578734 ± 0.009136**